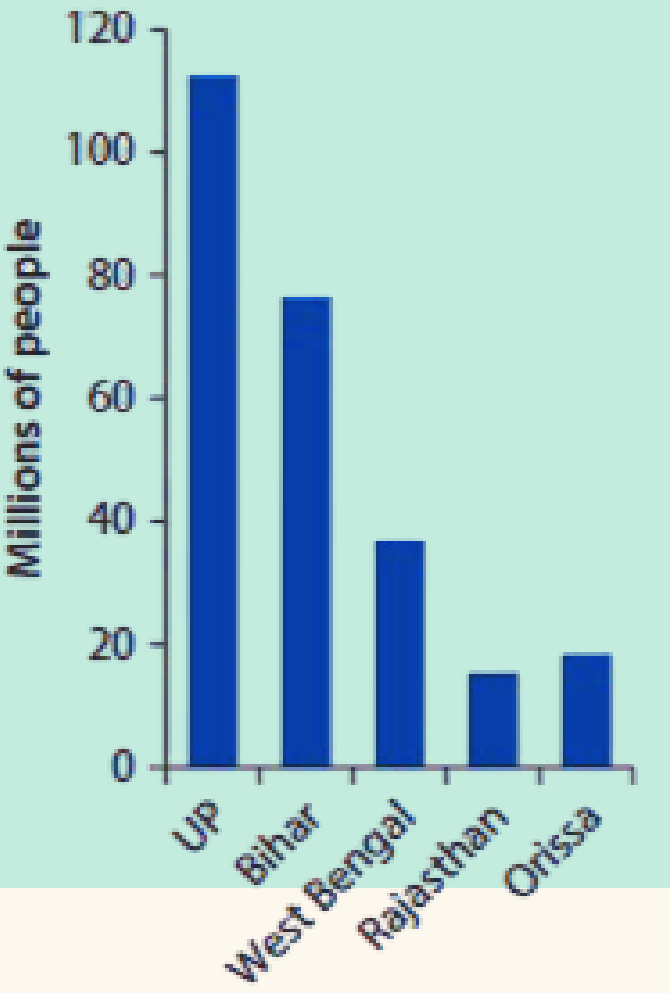


POWER FROM BELOW: DECENTRALIZED SOLAR MICROGRIDS FOR A JUST AND RESILIENT RURAL INDIA

Bakul U K

1 “Lights On, But Not for Everyone”

While India celebrates 100% village electrification, thousands of hamlets remain in the dark and invisible in policy, unreachable by power lines. “Over 230 million people in India still face unreliable or no electricity access (Banerjee et al., 2015). Grid extension is slow and costly in remote areas. This energy divide deepens social inequity.”

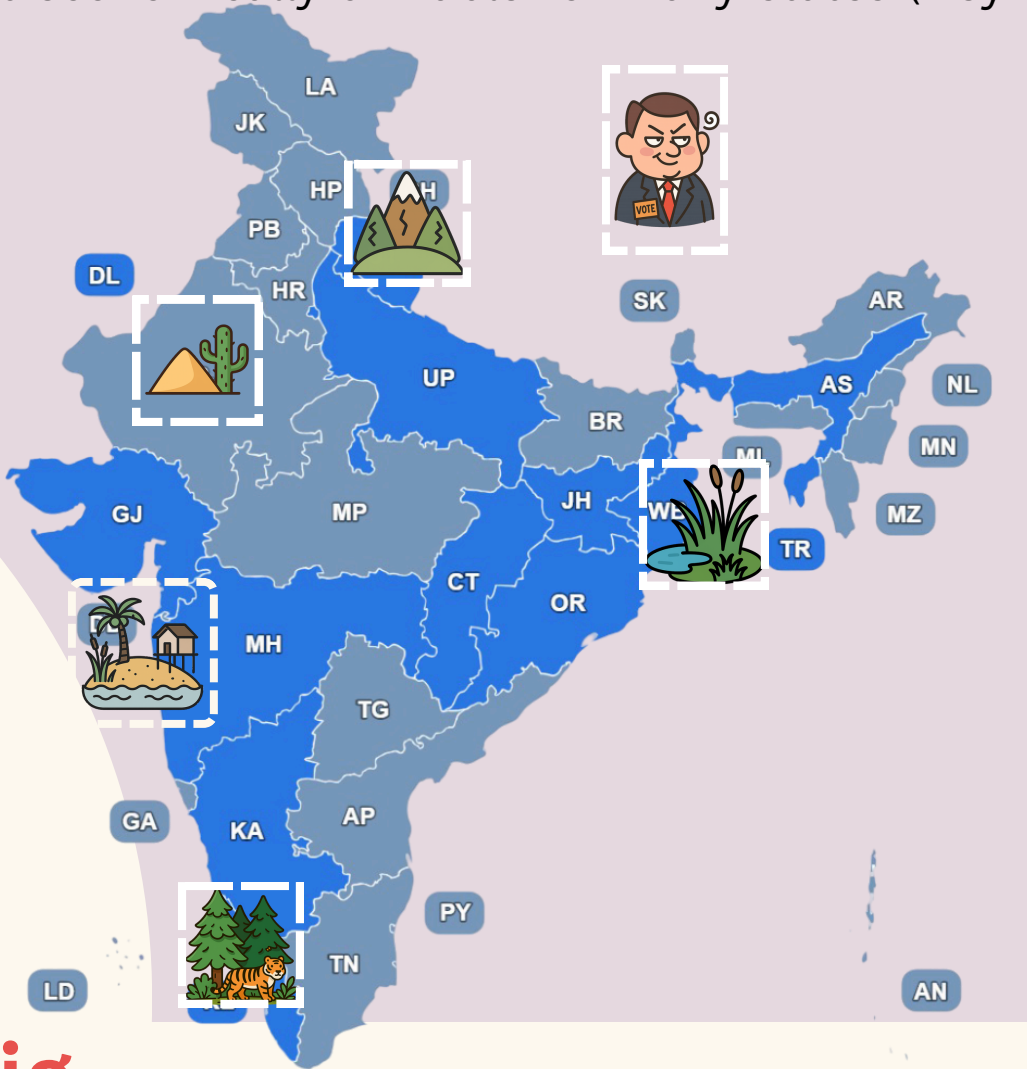


3 “How It’s Already Happening in India”

“India already has successful rooftop solar and mini-grid pilots in states like Gujarat, Maharashtra, and Uttar Pradesh. With state and central subsidies, over 1.2 GW of rooftop capacity has been installed (Gol, 2024).”

2 “Why the Grid Doesn’t Reach Them”

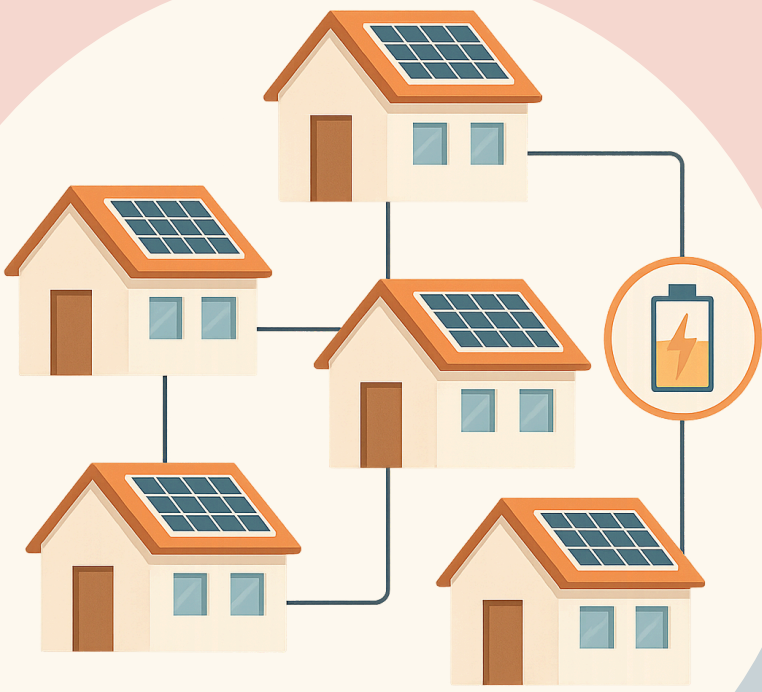
Mountain slopes, delta islands, and dense forests aren’t just far, they’re expensive, ecologically fragile, and politically neglected. moreover, “High cost of last-mile grid extension—up to ₹1.2 lakh/km—makes it economically unviable for many states (Dey et al., 2024).”



5 “Small Systems, Big Impact”

SHS empowers remote communities to run businesses, educate kids, and bounce back from disasters all without waiting for the grid.

“Each SHS offsets ~40L of kerosene/year, saving ~₹2,400 per household, while reducing CO₂ emissions by ~120 kg/year (World Bank, 2021).”

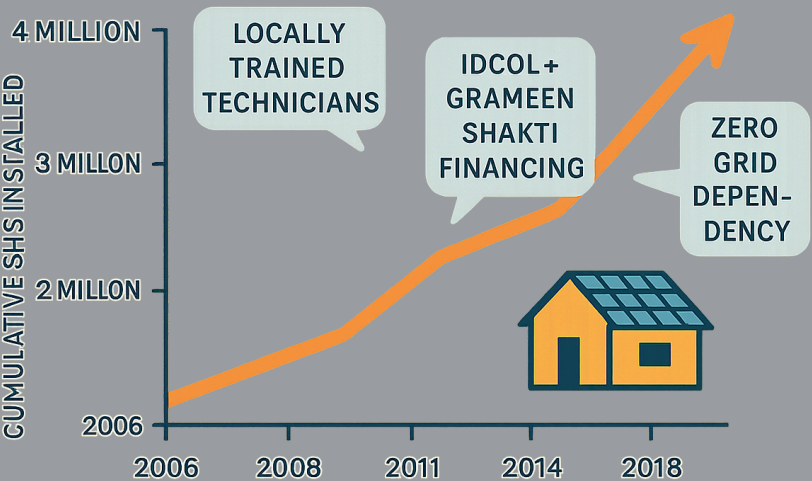


4 A Light from East: Bangladesh’s SHS

“From 2003–2018, Bangladesh installed 4+ million SHS, reaching 14% of its population—largely via microfinance and decentralized planning (Cabraal et al., 2021).”

A LIGHT FROM THE EAST

BANGLADESH’S SOLAR HOME SYSTEM (SHS) PROGRAM



6 “This Is About More Than Power”

This is about justice. It’s about people powering themselves — not being left behind. Climate action must begin from below.

“Decentralized solar solutions are not just backups, they are essential for climate-resilient, inclusive energy transitions. With falling panel costs and improved governance, India can lead a bottom-up renewable revolution.”